



Basic Principles of Medicine 2
Module: Endocrine & Reproduction | Lecture 12

Histology of the Male Reproductive System

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Recommended Reading

- Histology – A Text and Atlas Pawlina 9th Edition Chapter 22: Male Reproductive System Pg. 876-974

<https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences-lwwhealthlibrary-com.sgu.idm.oclc.org/book.aspx?bookid=3290>

Other Resources

- Mescher, A. L. (2018). The Male Reproductive System. In *Junqueira's Basic Histology Text & Atlas* (15th ed., pp. 439–459). McGraw Hill Education.
- Young, B., Woodford, P., & O'Dowd, G. (2013). *Wheater's functional histology* (6th ed.). Churchill Livingstone.
- Kumar, V., Abbas, A. K., & Aster, J. C. (Eds.). (2018). *Robbins basic pathology* (10th ed.). Elsevier.
- Tortora, G. J., & Nielsen, M. (2017). The Reproductive Systems. In *Principles of Human Anatomy* (14th ed., pp. 835–849). Wiley.

Objectives

SOM.MK.I.BPM2.2.ER.1.HCB.1077	Identify the locations of the testicular connective tissue coverings (tunica vaginalis, tunica albuginea).
SOM.MK.I.BPM2.2.ER.1.HCB.1078	Distinguish between the histological composition of the testicular connective tissue coverings (tunica vaginalis, tunica albuginea).
SOM.MK.I.BPM2.2.ER.1.HCB.1079	State the significance of the testicular connective tissue coverings (tunica vaginalis, tunica albuginea).
SOM.MK.I.BPM2.2.ER.1.HCB.1080	State the significance of testicular lobules.
SOM.MK.I.BPM2.2.ER.1.HCB.1081	Give the function of the mediastinum testis.
SOM.MK.I.BPM2.2.ER.1.HCB.1082	Describe the microscopic structure of the mediastinum testis.
SOM.MK.I.BPM2.2.ER.1.HCB.1083	List the functions of the seminiferous tubules
SOM.MK.I.BPM2.2.ER.1.HCB.1084	Describe the microscopic structure (including the major cell types of the epithelium) of the seminiferous tubules.
SOM.MK.I.BPM2.2.ER.1.HCB.1085	<i>Describe the major groups of testicular tumors (blue slide – independent review).</i>
SOM.MK.I.BPM2.2.ER.1.HCB.1086	Describe classical seminoma of the testis.
SOM.MK.I.BPM2.2.ER.1.HCB.1087	State the functions of the Sertoli cells.
SOM.MK.I.BPM2.2.ER.1.HCB.1088	Discuss the microanatomical features of Sertoli cells.
SOM.MK.I.BPM2.2.ER.1.HCB.1089	Describe the structure of the blood-testis barrier.
SOM.MK.I.BPM2.2.ER.1.HCB.1090	Identify Leydig cells by its microanatomical features and location.
SOM.MK.I.BPM2.2.ER.1.HCB.1091	State the function(s) of Leydig cells.
SOM.MK.I.BPM2.2.ER.1.HCB.1092	Distinguish between the microscopic structure of spermatogonia, primary spermatocytes, secondary spermatocytes and spermatids.

Includes reflective self-study questions for student use. Therefore, answers not provided.

Objectives

SOM.MK.I.BPM2.2.ER.1.HCB.1093	Describe the phases of spermatogenesis (including spermatogonial, spermatocyte, spermatid phases).
SOM.MK.I.BPM2.2.ER.1.HCB.1094	Give the function(s) of spermatozoa.
SOM.MK.I.BPM2.2.ER.1.HCB.1095	Describe the microscopic structure of spermatozoa.
SOM.MK.I.BPM2.2.ER.1.HCB.1096	State the effect(s) of Kartagener's syndrome on spermatozoa.
SOM.MK.I.BPM2.2.ER.1.HCB.1097	State the anatomical location of the seminal vesicles.
SOM.MK.I.BPM2.2.ER.1.HCB.1098	State the function(s) of the seminal vesicles.
SOM.MK.I.BPM2.2.ER.1.HCB.1099	Describe the microscopic structure of the seminal vesicles.
SOM.MK.I.BPM2.2.ER.1.HCB.1100	Identify the anatomical location of the prostate gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1101	State the function(s) of the prostate gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1102	Describe the microscopic structure (including corpora amylacea) of the prostate gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1103	Discuss the secretions of the prostate gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1104	Describe benign prostatic hyperplasia.
SOM.MK.I.BPM2.2.ER.1.HCB.1105	Describe adenocarcinoma of the prostate.
SOM.MK.I.BPM2.2.ER.1.HCB.1106	Give the anatomical location of the bulbourethral glands.
SOM.MK.I.BPM2.2.ER.1.HCB.1107	State the function of the bulbourethral glands.
SOM.MK.I.BPM2.2.ER.1.HCB.1108	Describe the microscopic structure of the bulbourethral glands.

Includes reflective self-study questions for student use. Therefore, answers not provided.

Male Reproductive System - Overview

1. Testes

- Seminiferous tubules
- Intratesticular ducts

2. Excurrent Ducts

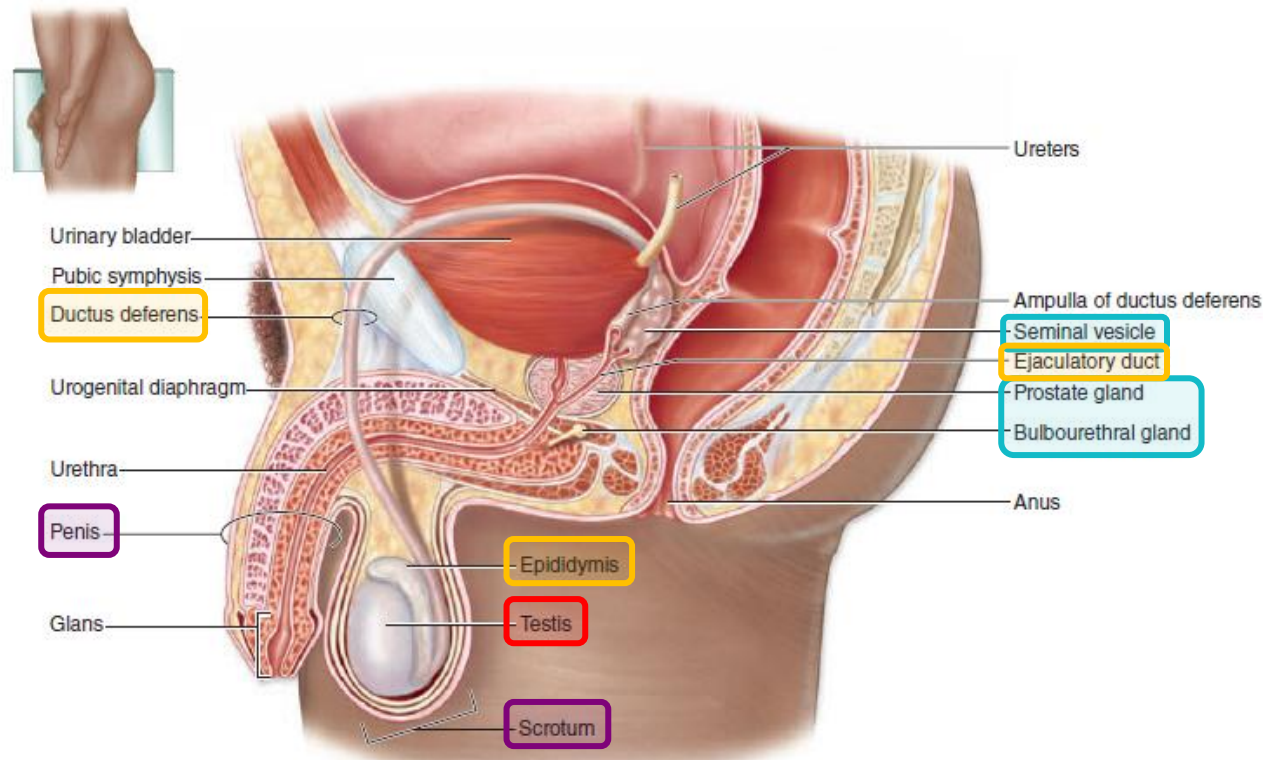
- Efferent tubules
- Epididymis
- Ductus deferens
- Ejaculatory duct

3. Accessory Glands

- Seminal vesicles
- Prostate gland
- Bulbourethral glands

4. External Genitalia

- Penis and urethra
- Scrotum



Testes: Coverings

1. Tunica Vaginalis *(derived from what embryological structure?)*

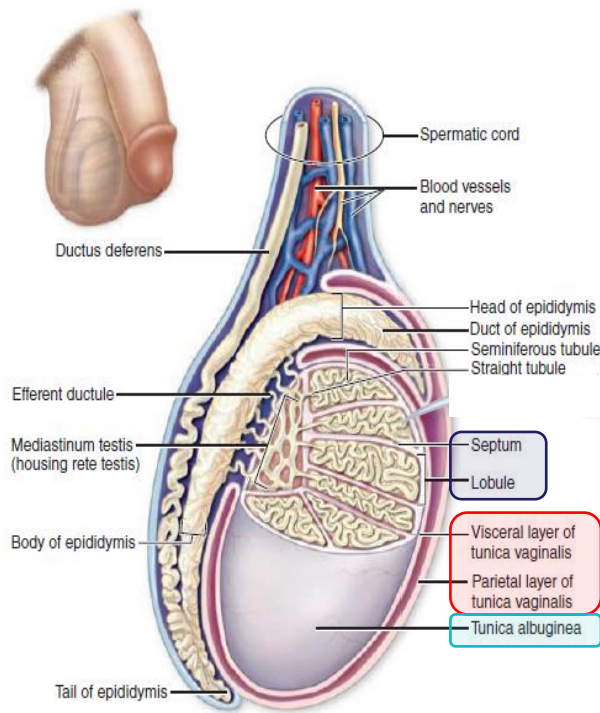
- Peritoneum *(how many layers?)*
- Covers anterolateral surface of testes

2. Tunica Albuginea

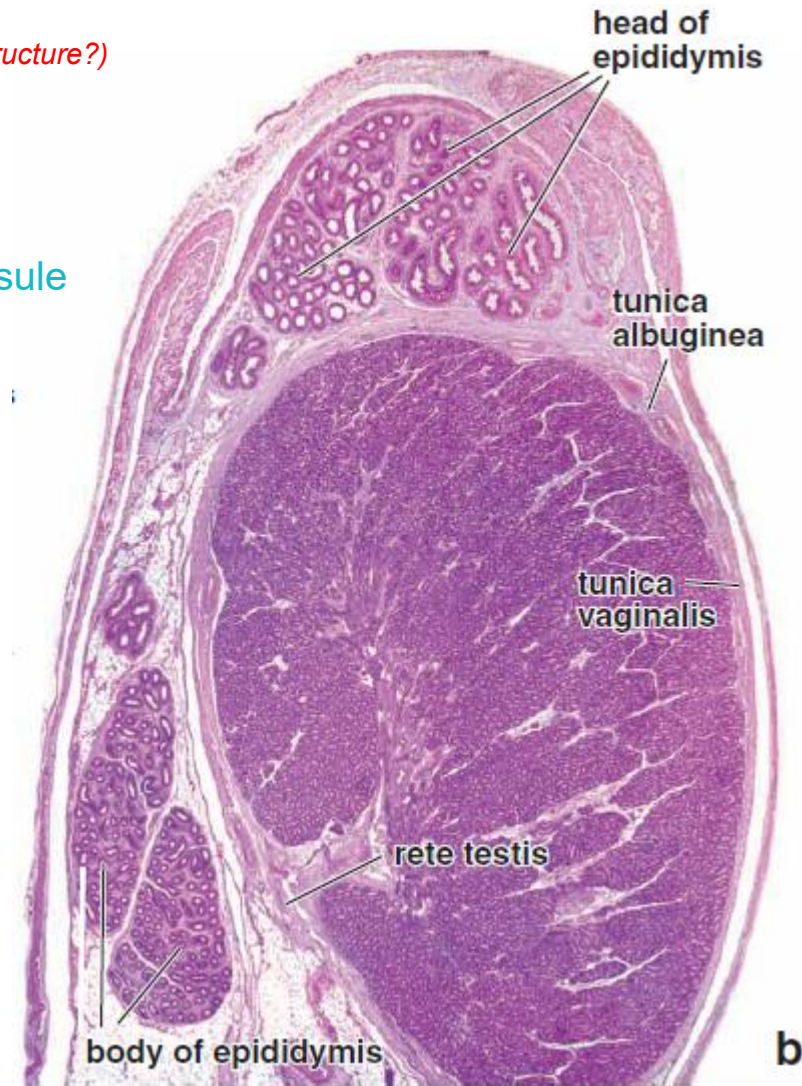
- Unusually thick dense connective tissue capsule
- Inner part: Tunica vasculosa
 - Loose CT containing blood vessels

3. Connective Tissue Septa

- Divides testes into ~250 lobules



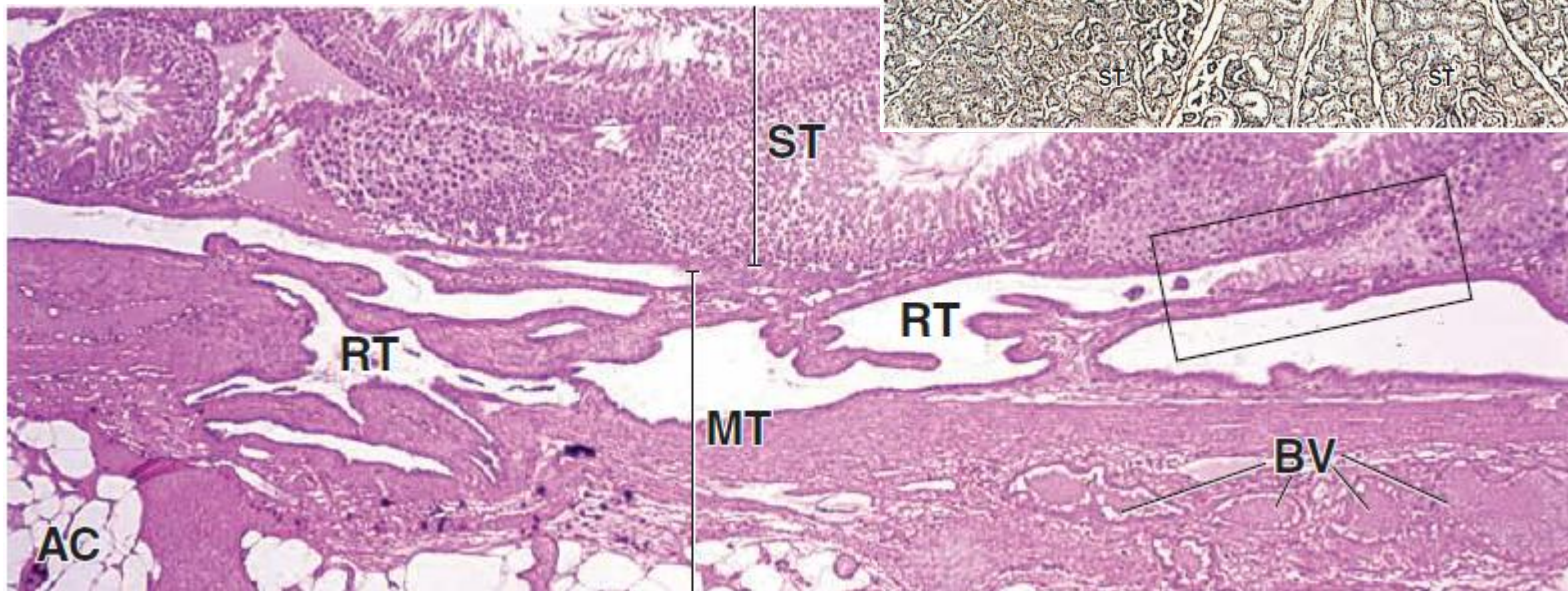
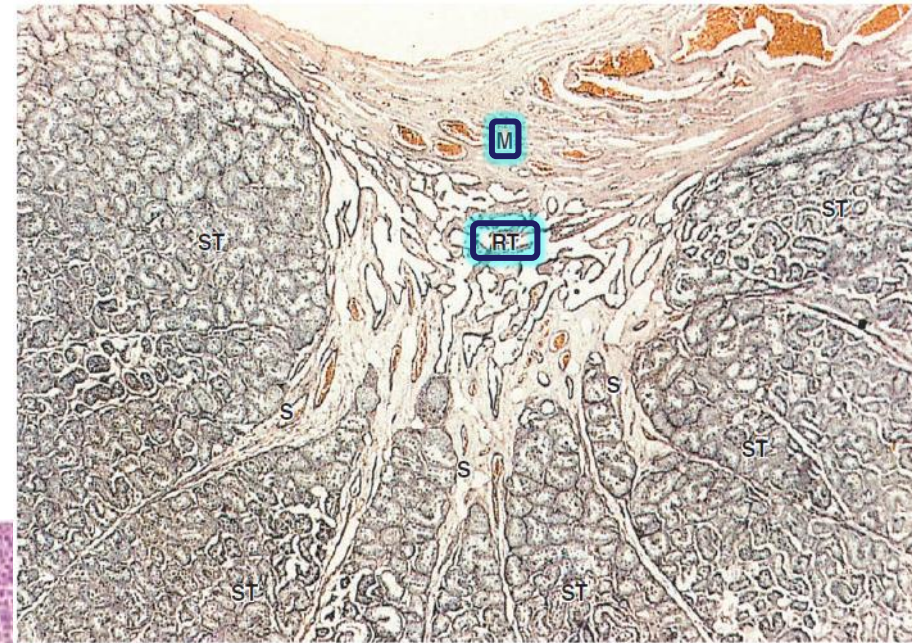
a



b

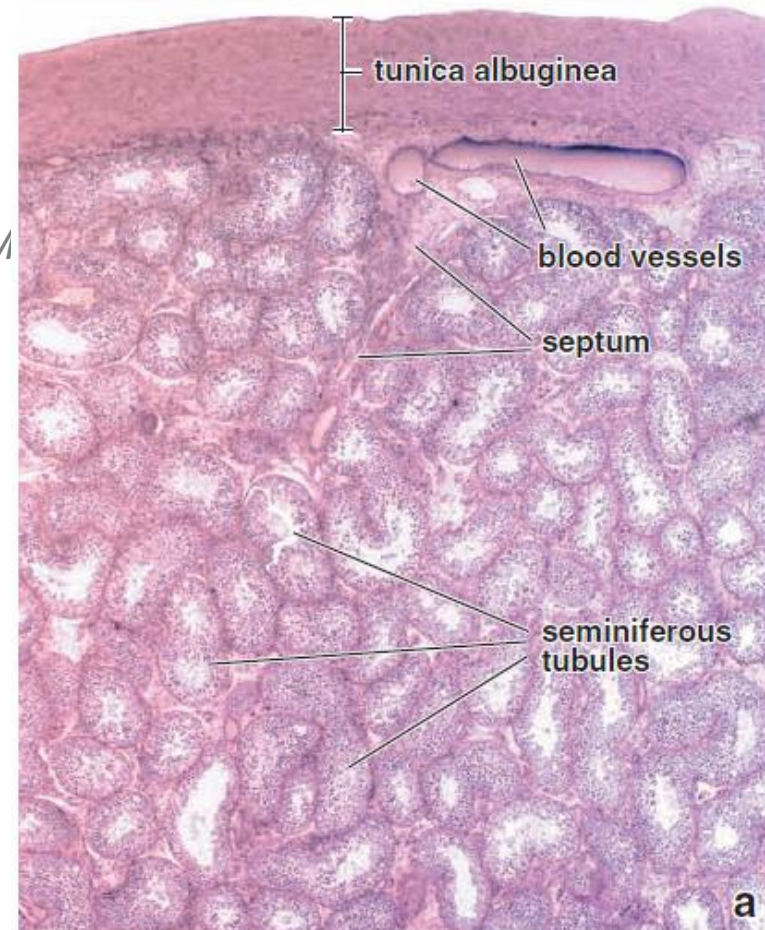
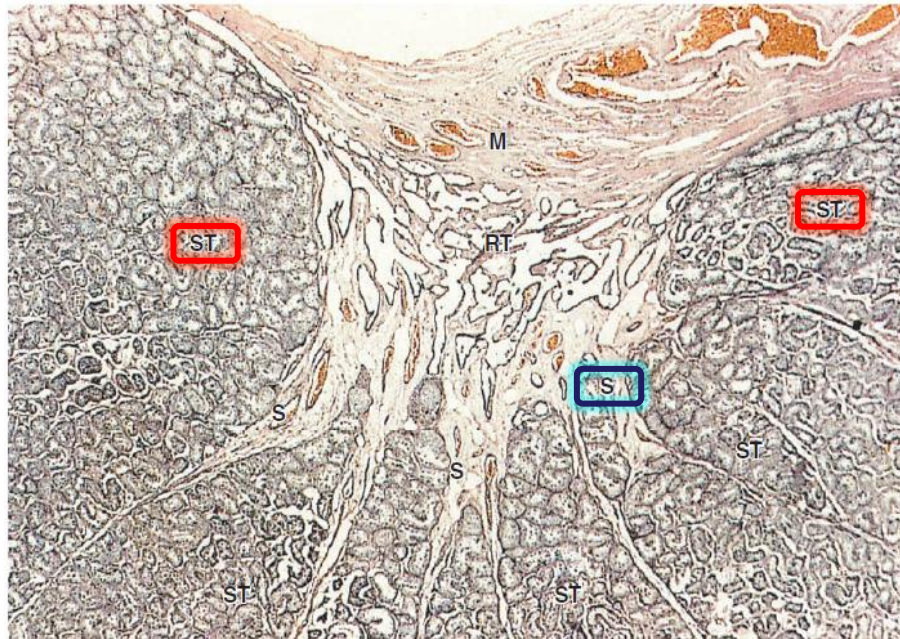
Mediastinum Testis

- Inward projection of thickened tunica albuginea (*M / MT*)
- Along posterior surface of testis
- Contents
 - Rete Testis (*RT*)
 - Blood vessels (*BV*)
 - Lymphatic vessels
 - Genital excurrent ducts



Testes: Lobules

- Bounded by incomplete connective tissue septae (S)
- **Parenchyma: Seminiferous tubules (ST)**
 - Several highly convoluted tubules (1-4 per lobule)
 - Produce sperm (spermatogenesis → Seminiferous epithelium)
- **Stroma: Interstitial/Leydig cells**
 - Produce testosterone (steroidogenesis)
- **Intratesticular ducts (see DLA)**
 - Straight tubules
 - **Rete Testes (RT in mediastinum testis – M*

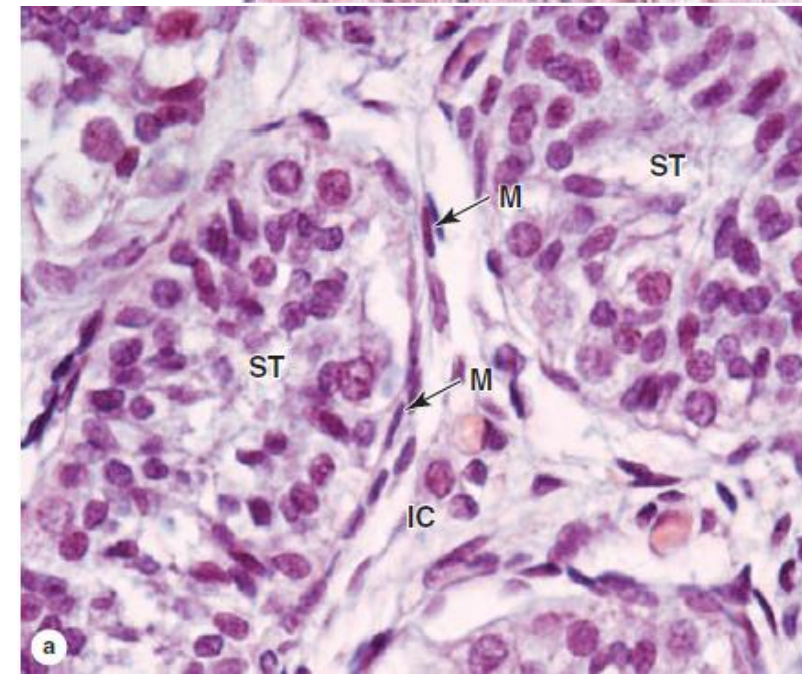
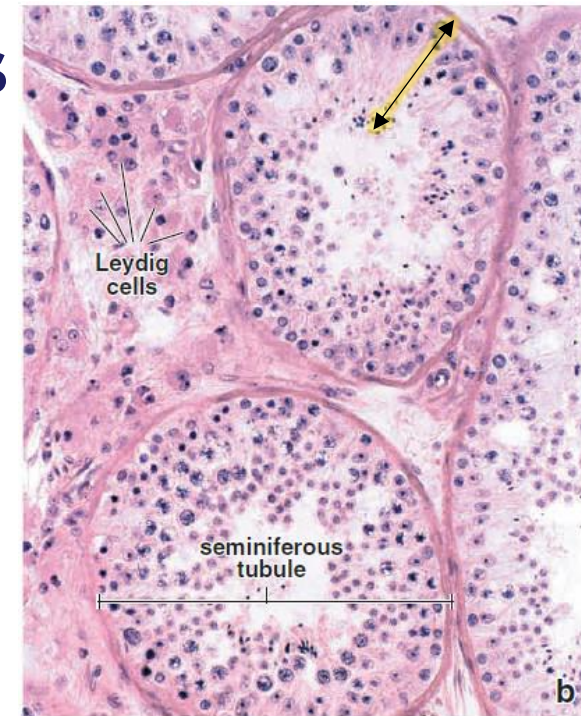


Testes: Seminiferous Tubules

[ST] ~50cm long convoluted tubule

Lined by seminiferous epithelium

- Complex stratified epithelium
 - **Cells**
 - **Sertoli cells**
 - **Spermatogenic cells**
- **Sits on tunica propria (lamina propria)**
 - *Also called 'peritubular tissue'*
 - Multilayered connective tissue – collagen fibrils
 - **3-5 layers peritubular contractile cells (myoid cells – M)**
 - **Actin filaments**: peristalsis – move spermatozoa + testicular fluid through tubules to ducts
 - **rER**: collagen synthesis
- Surrounded by stroma
 - Interstitial cells (*IC/ Leydig cells*)
 - Blood + lymphatic vessels

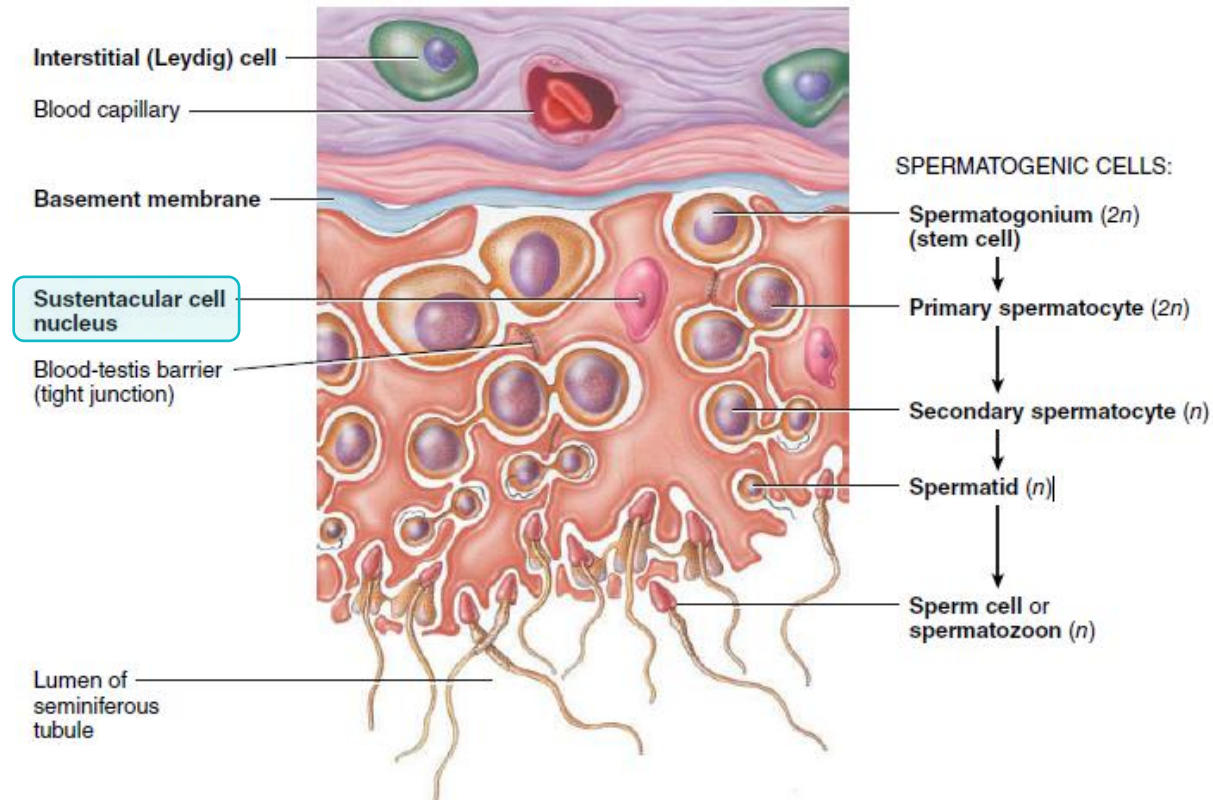


Seminiferous Epithelium: Sertoli cells

- Also called **Sustentacular or Supporting cells**
- **Surround spermatogenic cells (Sertoli cells → non-replicating after puberty)**
 - Support developing spermatozoa
- **Tall, irregular columnar cells**
 - Extend full thickness of epithelium
 - Extensive apical + lateral processes
- **Organelles**
 - Ovoid/triangular euchromatic nucleus
 - **Prominent nucleolus**
 - Extensive sER, well-developed rER
- Bound to each other by **unique junctional complexes** (*blood-testis barrier*)
- Phagocytose and break down residual bodies, abnormal spermatogenic cells

Secretions

- **Testicular Fluid**: facilitate passage of sperm through intratesticular ducts
- **Androgen-binding protein (ABP)**: binds testosterone (high levels)
- **Inhibin**: inhibits FSH secretion (endocrine*)
- Growth factors: e.g., **Müllerian-inhibiting substance (MIS)**



Seminiferous Epithelium: Spermatogenic cells

1. Spermatogonia (*most immature cells – basal lamina*)

- Stem cells ($2n$, $2d$)
- **Type A dark (Ad)**: reserve
- **Type A pale (Ap)**: renewing
- **Type B**: *differentiate into primary spermatocyte*

- Regularly replicate and differentiate into mature sperm
- Organized in poorly defined layers of progressive development
- Between adjacent Sertoli cells

2. Spermatocyte

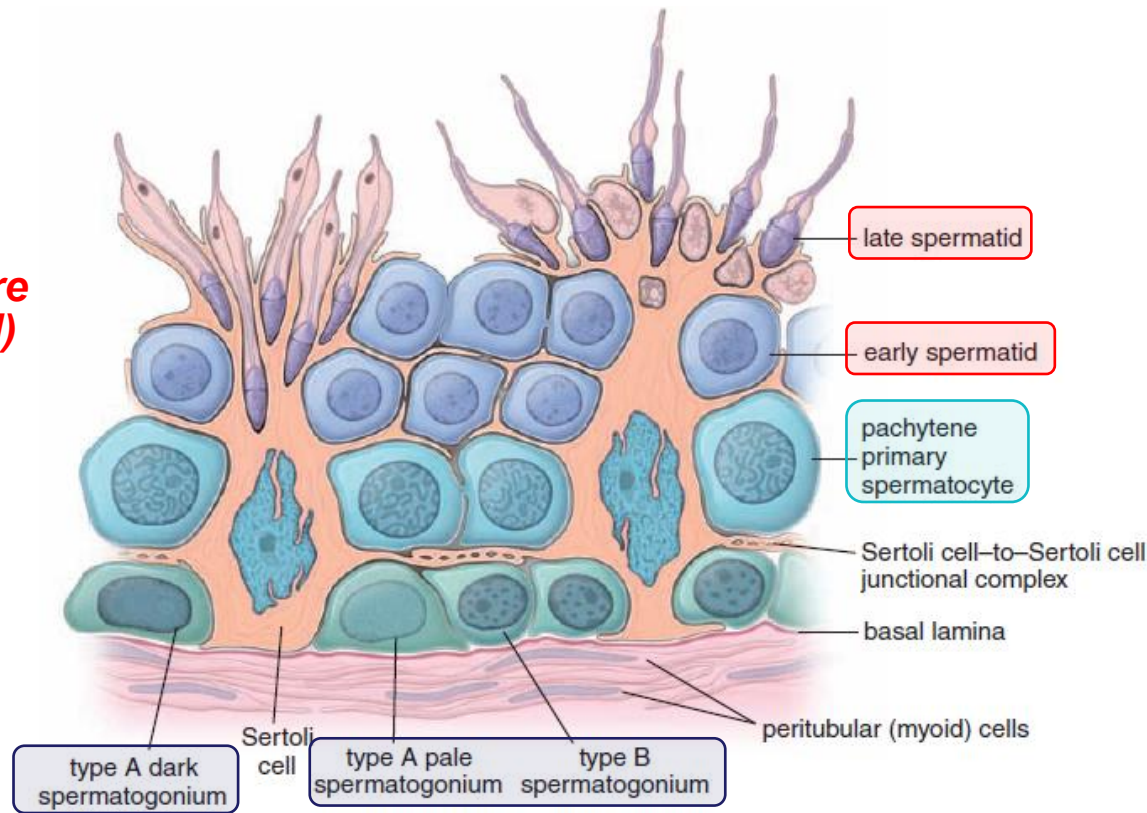
- **Primary** ($2n$, $4d$)
- **Secondary** ($1n$, $2d$)

3. Spermatid (*most mature cells – apex of Sertoli cell*)

- **Early** ($1n$, $1d$)
- **Late** ($1n$, $1d$)

✓ Spermatozoa ($1n$, $1d$)

❖ Fertilize ova



Seminiferous Epithelium: Spermatogenic Cells

1. Spermatogonia: oval nuclei

- **Type A dark (Ad)** – intensely basophilic chromatin
- **Type A pale (Ap)** – lightly staining chromatin
- **Type B (B)** – **spherical**, pale nucleus with large chromatin clumps

2. Primary spermatocytes (Ps)

- Largest cells, large coarse clumps or threads of chromatin
 - Meiosis I – days

3. Secondary spermatocytes

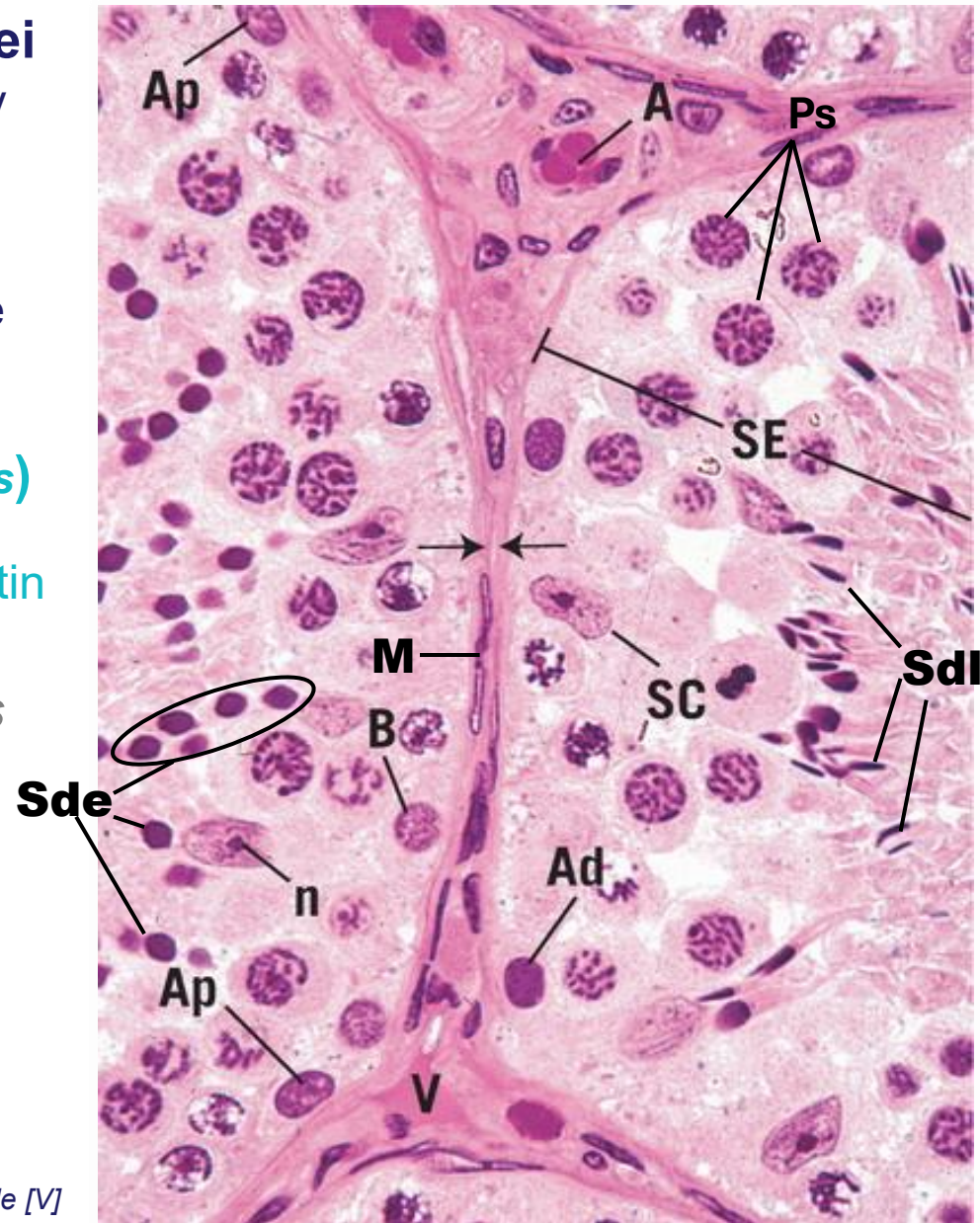
- *Rarely seen (not shown)*
 - Meiosis II – hours

4. Early Spermatids (Sde)

- Smaller cells; small dense, round nuclei

5. Late Spermatids (Sdl)

- Dense, elongated nuclei



Myoid cell [M], Sertoli cell nucleus [SC], Arteriole [A], Venule [V]

Practice Slide

Label the following cells – include justification for labels

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____



Phases of Spermatogenesis

1. Spermatogonial Phase (Mitosis)

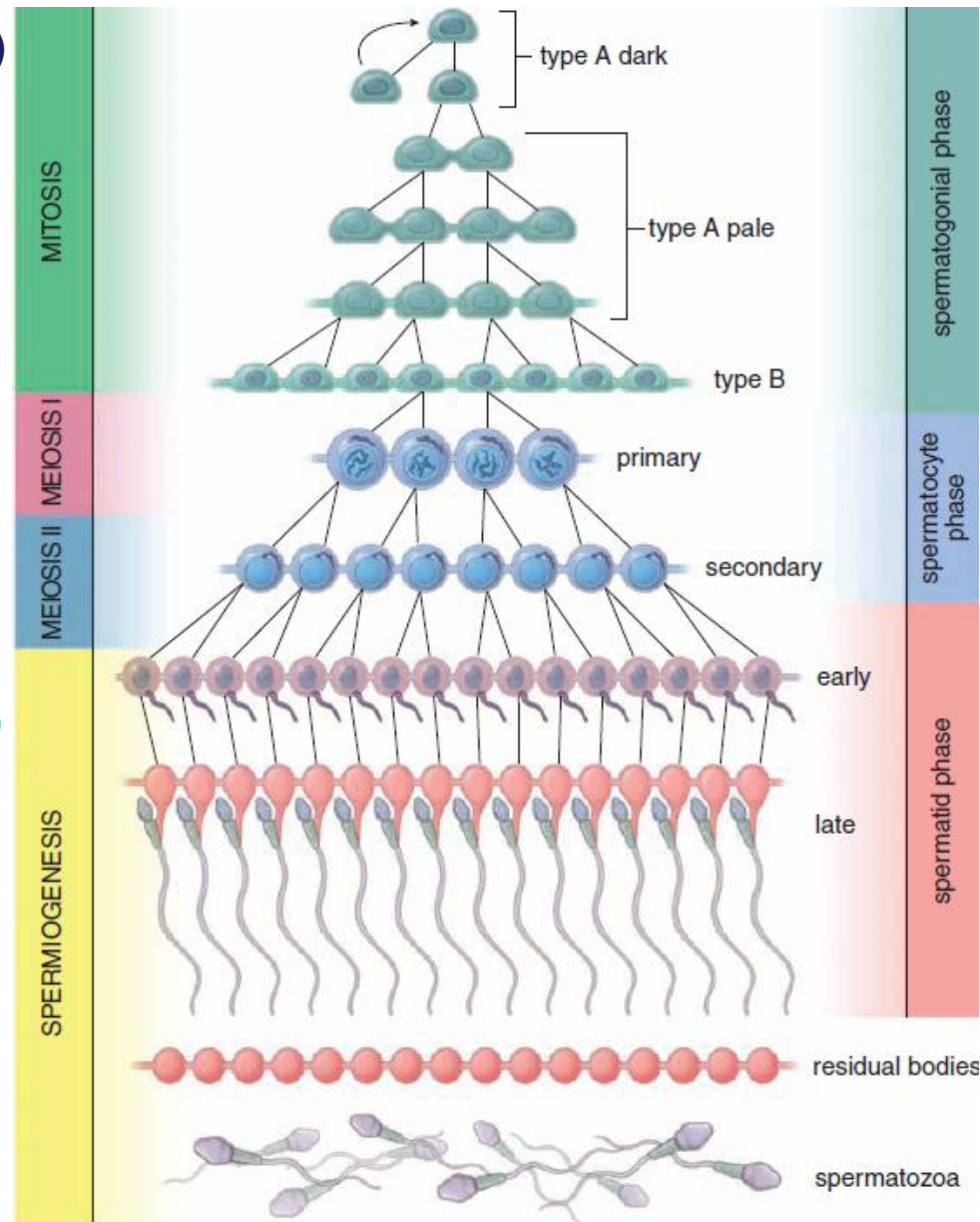
- Replicate self and provide committed population ($2n$, $2d$)
- Type B is last of this phase → primary spermatocyte
- *All Ap progeny connected by cytoplasmic bridge (synchronous development)*

2. Spermatocyte Phase (Meiosis)

- Reduction in chromosome number and DNA amount
- **Primary spermatocyte:** ($2n$, $4d$)
 - Undergoes Meiosis I
- **Secondary spermatocyte:** ($1n$, $2d$)
 - Undergoes Meiosis II

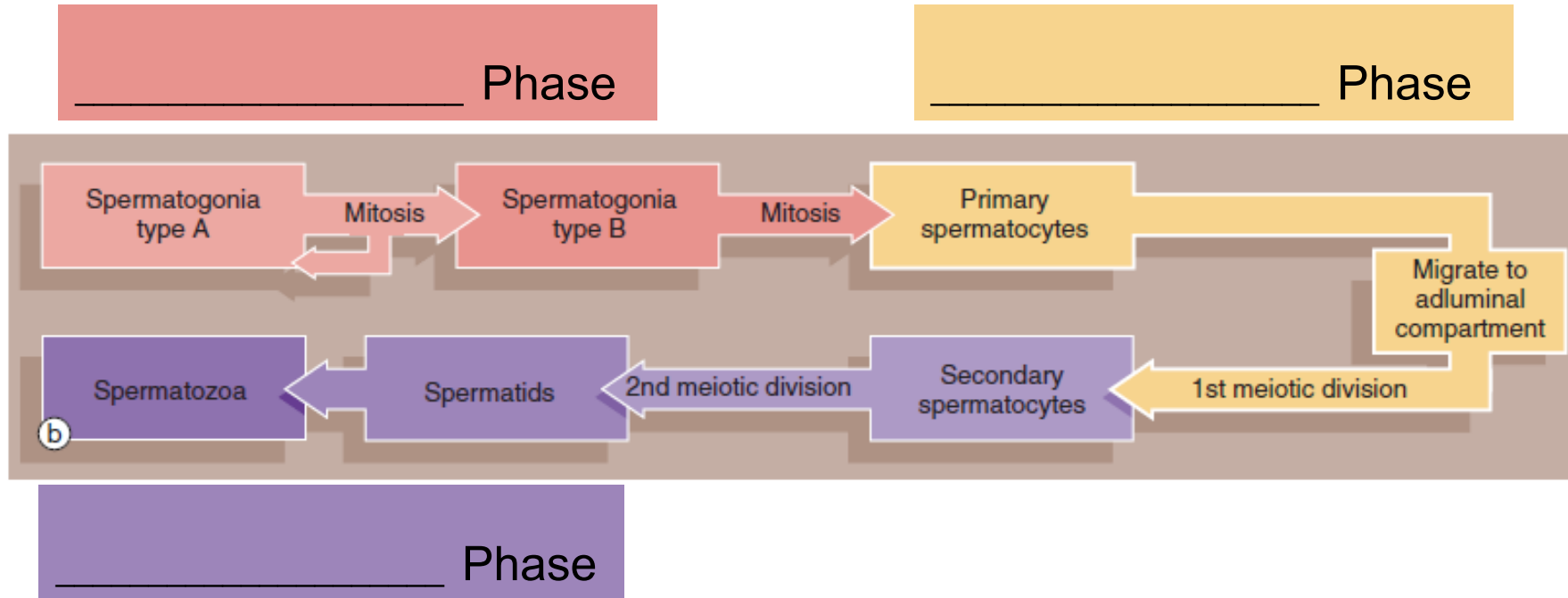
3. Spermatid Phase (Spermiogenesis)

- Remodeling + maturation
- Spermatid → spermatozoa
- ($1n$, $1d$)



Spermatogenesis

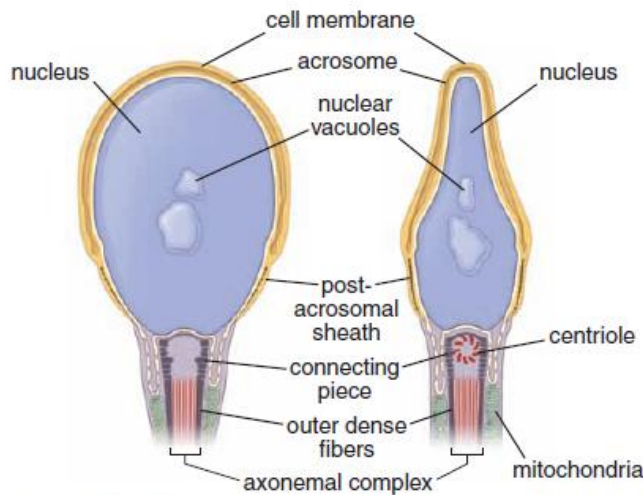
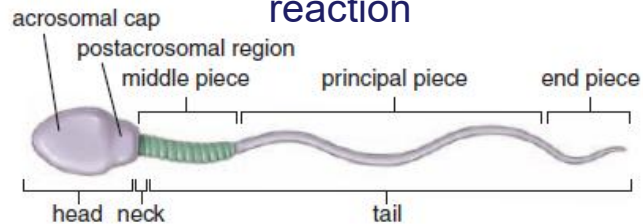
Label the phases.



Spermatozoa

Head (flattened and pointed)

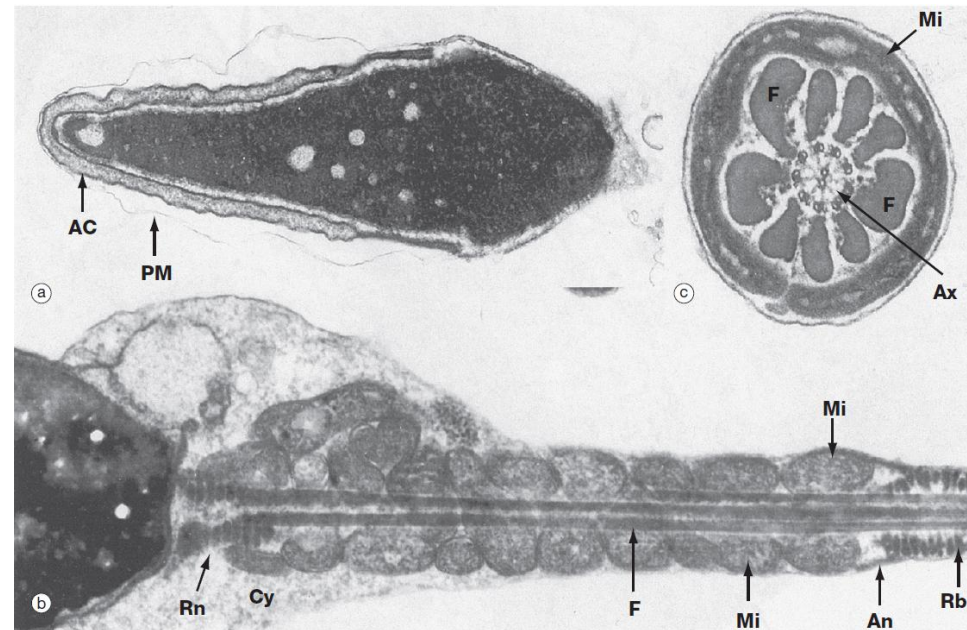
- **Acrosomal cap (AC)**
 - Covers anterior 2/3's of **nucleus**
 - Contains **acrosomal enzymes** (penetrate ovum)
 - E.g., hyaluronidase, neuraminidase
 - Enzyme release → acrosomal reaction



HEAD AND NECK REGION

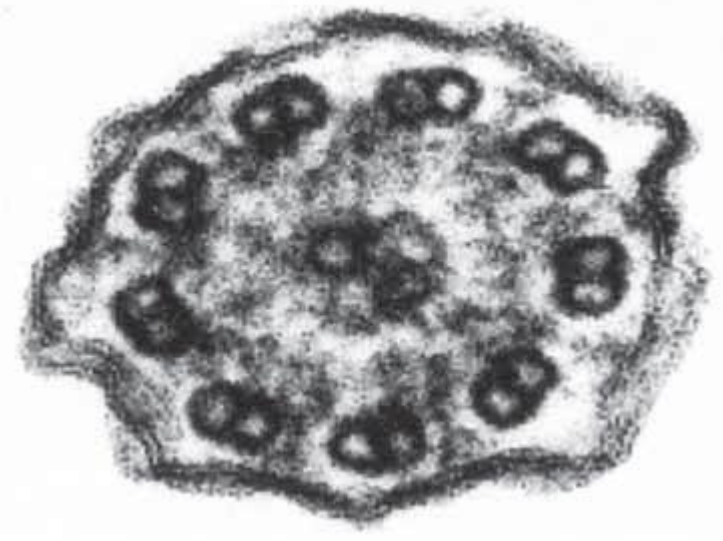
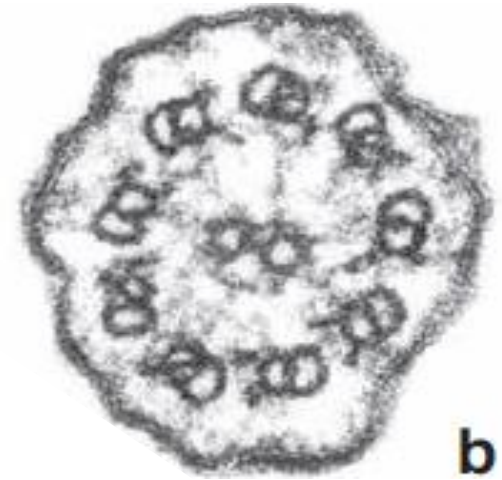
Tail (Flagellum)

- **Neck:** short, centrioles, origin of coarse fibers (F)
- **Middle piece:** helical mitochondria (Mi) sheath, (F), **axonemal complex (Ax)**
 - Responsible for sperm motility
- **Principal piece:** fibrous sheath (fibrous ribs – Rb) external to (F), (Ax)
- **End piece:** axonemal complex only



Kartagener's Syndrome

- Structural abnormality in ciliogenesis
 - **Result in absence of dynein arm**
 - Abnormal axoneme formation (9+2)
 - → mucociliary transport failure
- **Misorientation in basal feet**
- Impairment in all systems with cilia (e.g., respiratory, reproductive, etc.)
- **Male structures with 9+2 arrangement**
 - Sperm flagellum, cilia in efferent ductules
 - Infertility due to non-motile flagella
- Collective name for several genetic disorders with affected ciliogenesis:
 - Primary Ciliary Dyskinesia (PCD) or
 - Immotile Cilia Syndrome



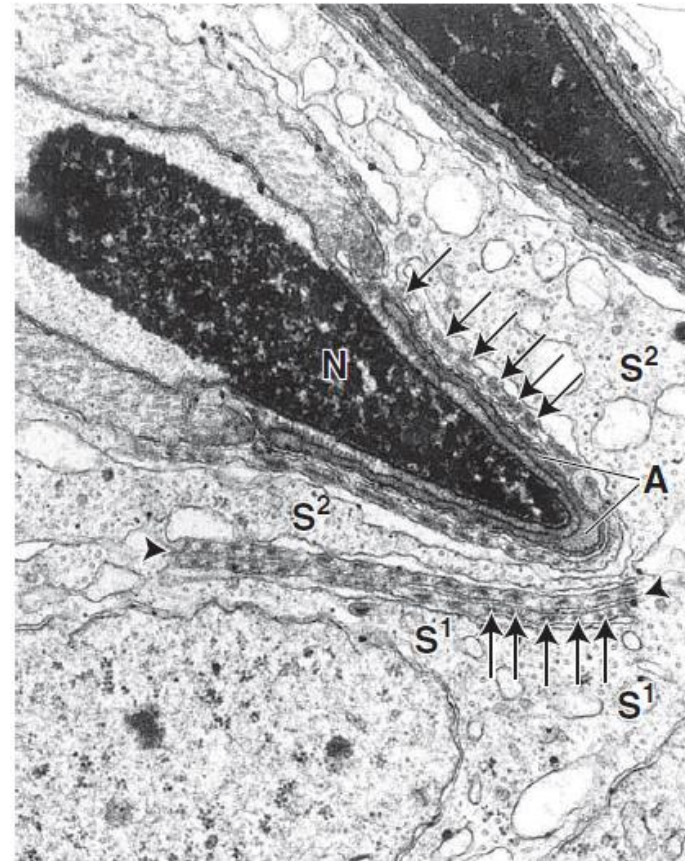
Blood-Testis Barrier

Sertoli cell-to-Sertoli cell junctional complex

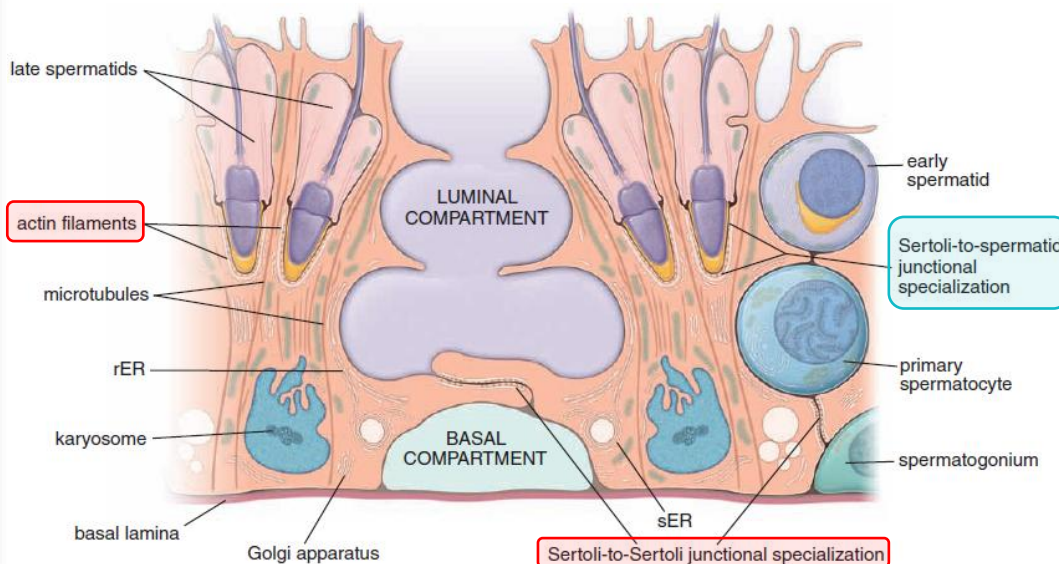
- Exceedingly tight junction (zona occludens)
- Flattened sER cisterna, Actin filament bundles
- **Divides epithelium**
 - **Basal compartment:** premeiotic
 - **(Ad)Luminal compartment:** meiotic + postmeiotic
- **Site of blood-testis barrier**
 - Controls fluid composition, no plasma cells/antibodies
 - Isolates '*antigenic*' haploid germ cells from immune system

Similar complex between Sertoli cell and spermatid

- No tight junction, spermatid lacks flat sER and actin bundles

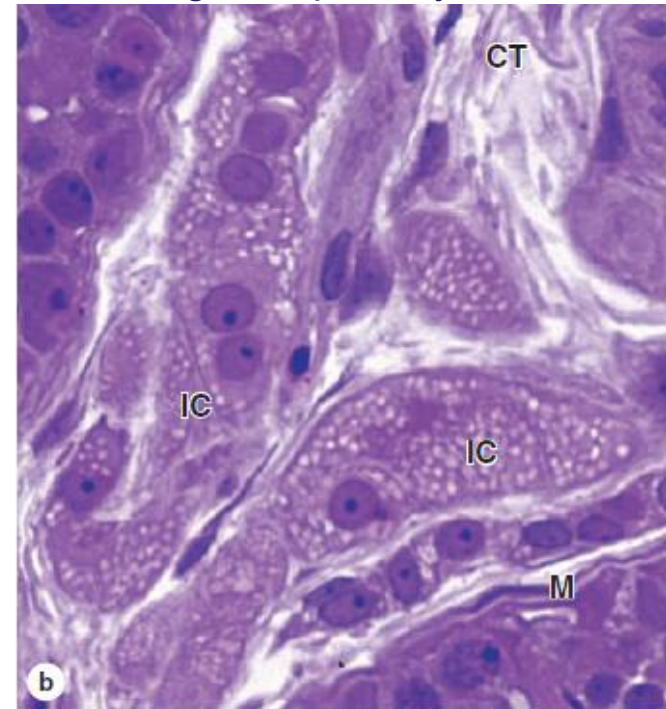
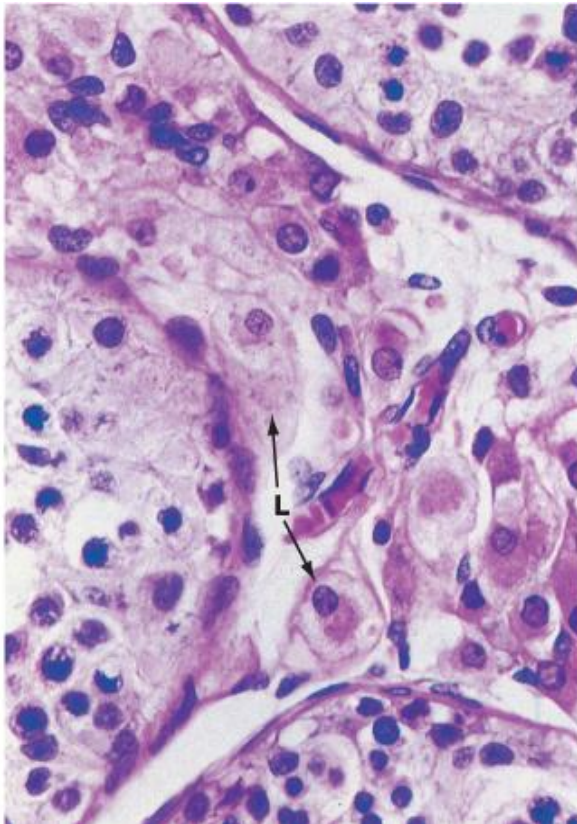


S¹: one Sertoli cell
S²: adjacent Sertoli cell
A: acrosome
N: nucleus
↑: actin bundles
➤: limit of junction

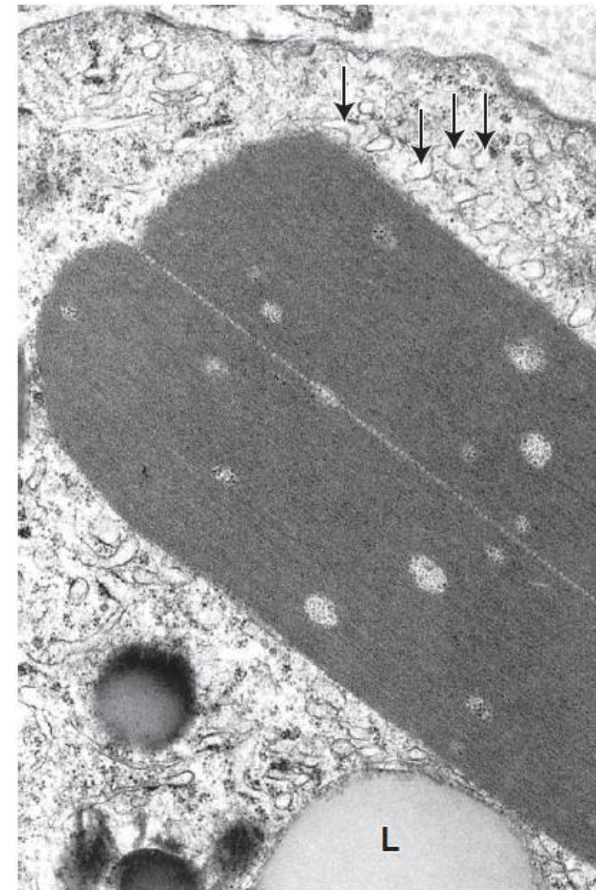
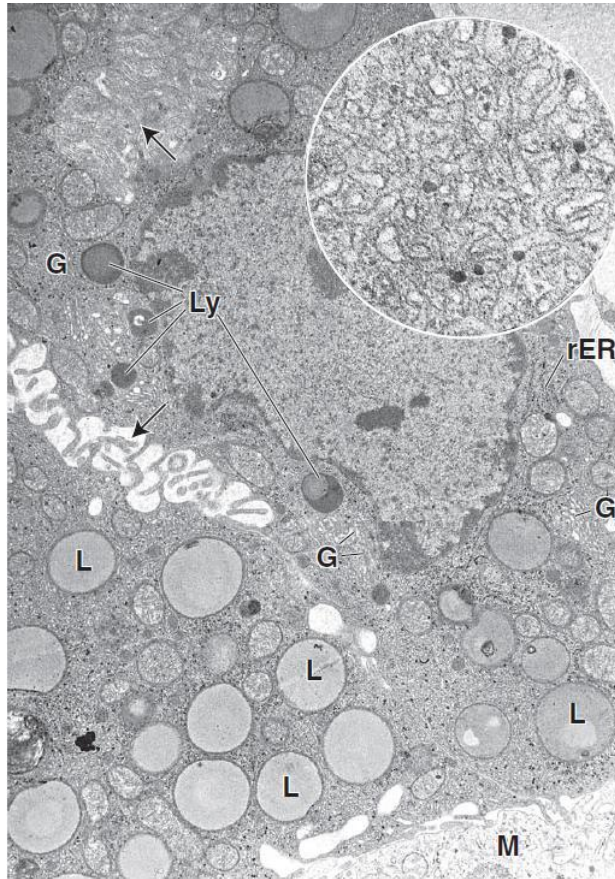


Leydig Cells

- Also called '*Interstitial cells*' (IC)
- Located in stromal spaces between adjacent seminiferous tubules
- **Large, polygonal, eosinophilic cells with small, round nuclei and numerous lipid droplets**
- **Functions:**
 - Secrete **testosterone**, **androgens**, **insulin-like protein 3 (INSL3)**, **oxytocin**
 - **Embryo:** normal testis development + descent
 - **Puberty:** spermatogenesis, accessory gland secretion, secondary sex characteristics
 - **Adult:** maintenance of changes begun in puberty



Leydig Cells



- **Steroid hormone producing cell**
 - L – Lipid droplet, G – Golgi
 - sER (inset → eosinophilia)
 - rER, M - Macrophage
 - Ly – Lysosome
 - Microvilli (arrows)
- **Reinke crystal:**
 - rod-shaped cytoplasmic crystals (unknown function)
 - ↓ - sER, L – Lipid droplets

Review Slide – Big picture (Part 1)

The parenchyma of the testicles are the _____, which are responsible for _____. The surrounding stromal tissue is responsible for _____. The _____ or _____ cells of the stroma are responsible for producing the hormone _____ that is crucial for the normal development of sperm.

The _____ tubule is lined by _____ epithelium comprised of _____ or _____ and _____ cells. The _____ cells support the _____ cells as they develop into mature _____. The _____ secreted by the _____ cells maintains the high _____ concentration needed for sperm production. These cells also produce _____, _____ and _____.

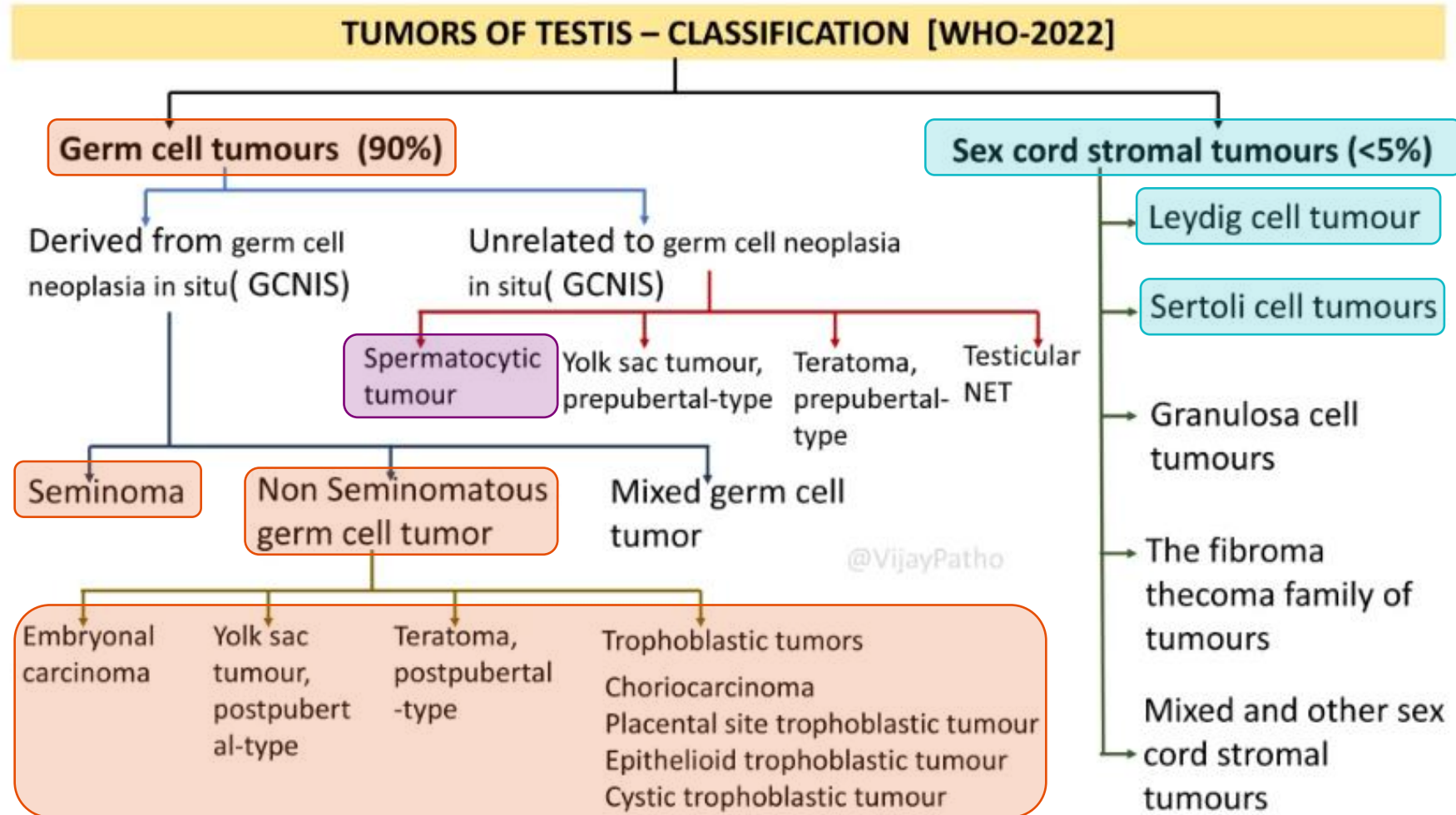
The most immature spermatogenic cells are known as _____. There are three types, Type _____ which are referred to as the reserve stem cells, Type _____ which are the _____ stem cells and _____ which are committed to differentiating into _____. _____ cells are the last cells in this phase. All stem cells undergo the process of _____ during the _____ phase of spermatogenesis.

Subsequently, during the _____ phase, _____ and _____ undergo the process of _____, resulting in a reduction in the number of chromosomes and amount of DNA. Leading up to this, _____ (2n, 2d) cells undergo _____ to form _____ (2n, 2d), which then undergo _____ to produce _____ (2n, 2d) which finally undergo the last phase of _____ to form _____ (2n, 2d).

During the final phase called _____ or the _____ phase, _____ remodel and mature into _____ (2n, 2d) – which are responsible for _____. These are comprised of a _____ that is attached to the apex of the _____ cell and a _____ that extends into the lumen of the _____. The _____ contains the _____ and the _____ which contains enzymes responsible for the _____ during fertilization. The _____ consists of four parts, the _____, _____, _____ and _____ pieces.

Testicular Tumors – Independent Review

WHO Classification of Testicular Tumors



Focus on highlighted groupings – most helpful for Year 2 (Pathology)

Classical Seminoma

- **Presentation**

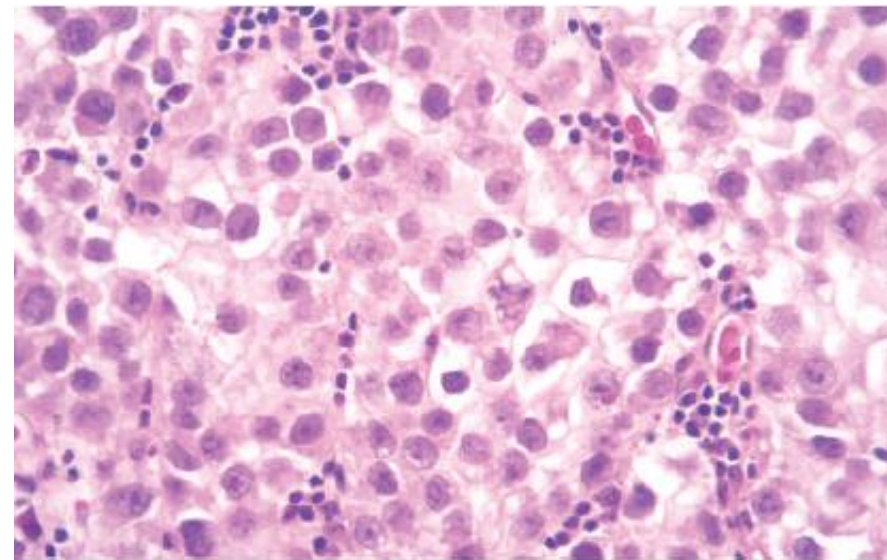
- Soft, well-demarcated, gray-white tumor
- Bulges from cut surface of affected testis
- Elevated serum human chorionadotropin (hCG) – 10%

- **Macroscopic changes (gross inspection)**

- Solid, homogenous, fleshy mass with distinct membrane
- Large, round, nodular/lobular, pale-tan to yellow-cream

- **Microscopic changes**

- Sheets of large, uniform, polygonal cells with distinct cells borders
- Clear, glycogen-rich cytoplasm, round nuclei, prominent nucleoli
- Poorly divided lobules with fibrous septa & lymphocyte infiltration



Accessory Sex Glands

Seminal Vesicles (Paired)

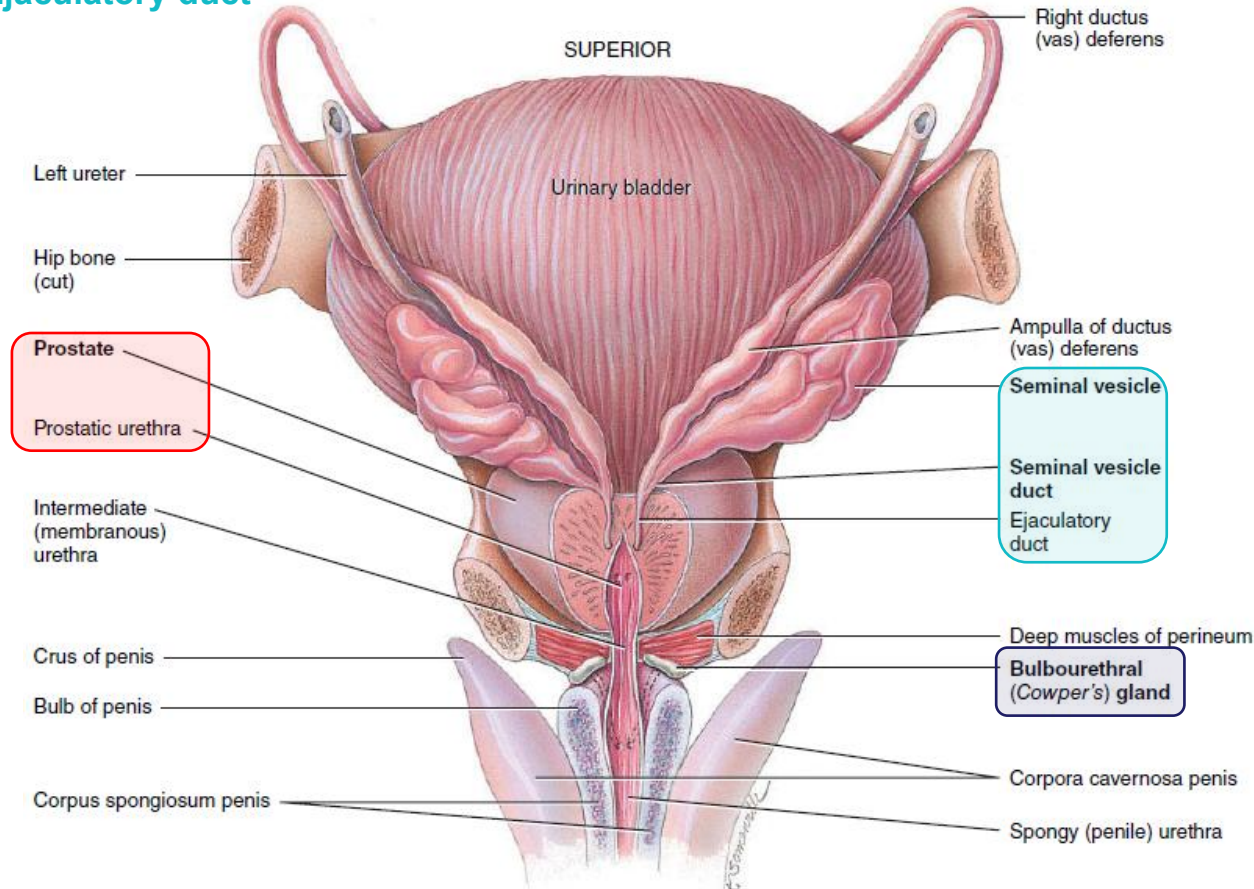
- Elongate, highly folded, tubular
- Posterior wall of bladder
- Excretory duct:
 - Short, narrow, straight
 - Joins ampulla of ductus deferens
 - → **Ejaculatory duct**

Prostate Gland

- Largest accessory sex gland
- Single gland inferior to bladder
- Size + shape ~ golf ball/walnut
- Surround prostatic urethra

Bulbourethral Glands (Paired)

- Also called 'Cowper's glands'
- Pea-sized; inferior to prostate
- Either side of membranous urethra
- Duct opens into penile (spongy urethra)



Accessory Sex Gland: Seminal Vesicle

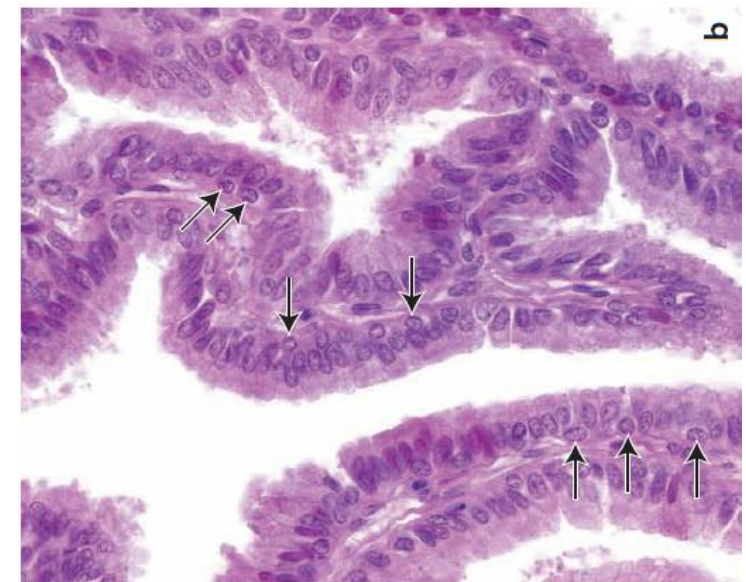
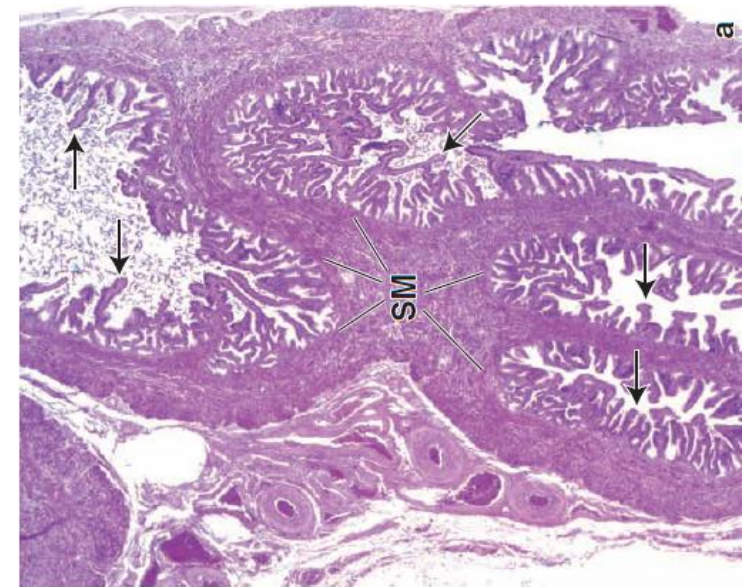
Mucosa

- **Pseudostratified columnar epithelium**
 - Tall non-ciliated columnar cells
 - Well-developed rER, secretory vacuoles
 - Small, round basal cells (*arrows in b*)
 - Stem cells (sparse)
 - Sit on loose CT lamina propria
- **Extensive folds/ridges (*arrows in a*)**
 - Only one lumen – all spaces connected
- **↑ surface area for secretion**
 - Whitish yellow, alkaline viscous material
 - **Fructose** (ATP production by sperm)
 - Prostaglandins
 - Produces ~65-75% volume of semen
 - Under control of testosterone

Thin Smooth muscle (SM) layer

- Inner circular and outer longitudinal layers
- Contraction during ejaculation
 - Discharges secretions
 - Flush sperm out of urethra

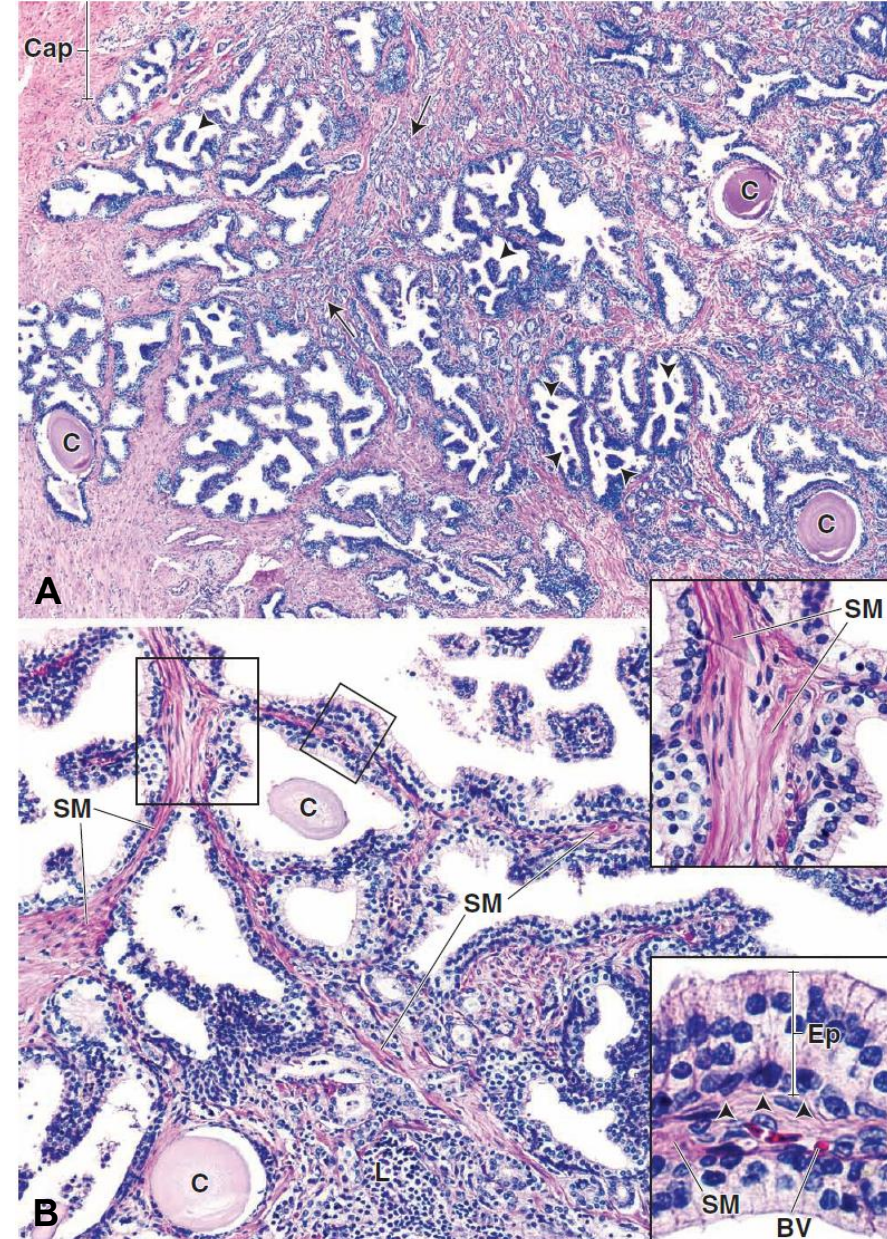
Fibrous coat



Accessory Sex Gland: Prostate Gland

30-50 tubuloalveolar glands

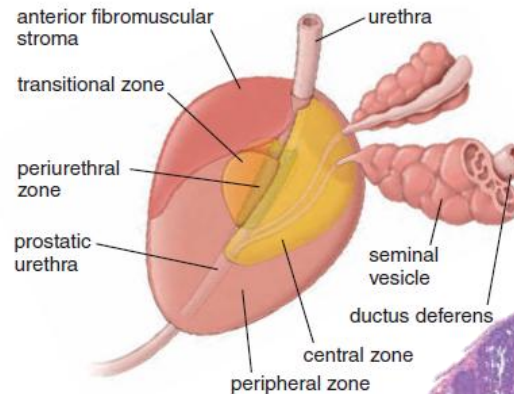
- **Three concentric layers**
 1. **Inner mucosal layer**
 - Secrete directly into urethra
 2. **Intermediate submucosal layer**
 - Secrete via ducts (prostatic sinus)
 3. **Peripheral layer**
 - Contains main prostatic glands
 - Secrete via ducts (prostatic sinus)
- **Glandular epithelium (Ep)**
 - Simple or pseudostratified columnar
 - Uneven contour: epithelial islands (*arrowheads in A*)
- **Corpora amylacea (C)**
 - Prostatic concretions (↑ w/ age)
 - Aggregates of dead epithelial cells + precipitated secretions (may calcify)
 - Concentric lamellar appearance
- **Anterior fibromuscular stroma**
 - Dense irregular connective tissue
 - Large amount of smooth muscle (SM)
 - Contract during ejaculation
- **Fibroelastic Capsule (Cap)**
 - Contain smooth muscle
 - Septa make lobules



Accessory Sex Gland: Prostate Gland

1. Central zone (CZ)

- Surrounds ejaculatory ducts (ED)
- 25% glandular tissue
 - **Submucosal glands** (*which layer?*)
- More prominent, slightly basophilic cytoplasm, larger nuclei



Cap: capsule
Sp: fibrous septa
St: fibrous stroma

2. Peripheral zone (PZ)

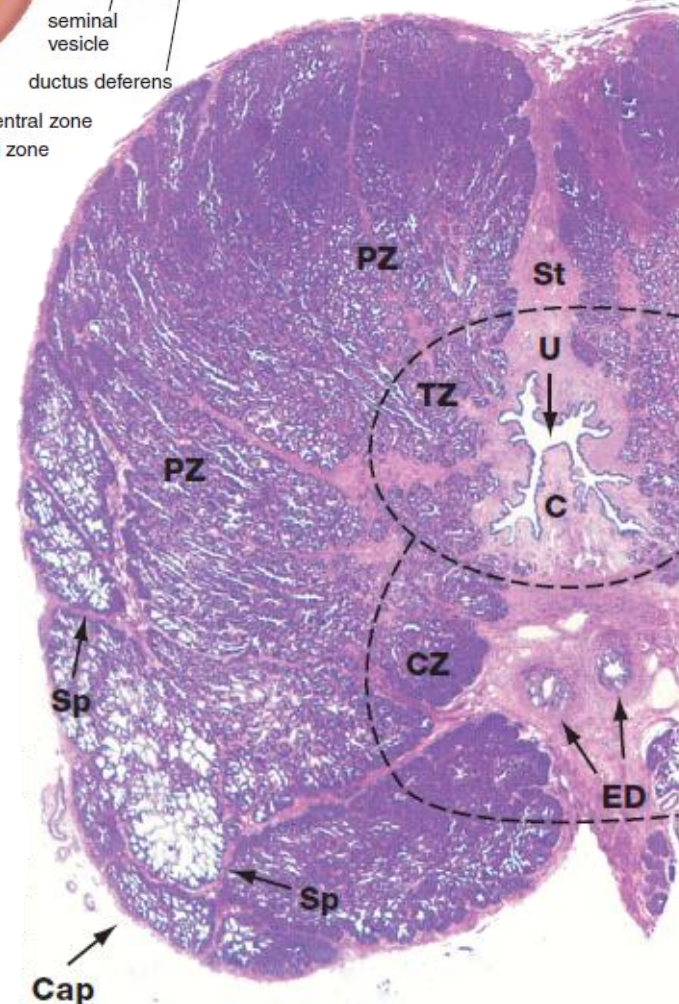
- Surrounds central zone
- Occupies posterior + lateral gland regions
- 70% glandular tissue
 - **Main glands** (*which layer?*)
- **Clinical correlate:** Most prostatic carcinomas arise here (DRE)

3. Transitional zone (TZ)

- Surrounds prostatic urethra (U)
- 5% glandular tissue
 - **Mucosal glands** (*which layer?*)
- **Clinical correlate:** Benign prostatic hyperplasia (BPH)

4. Periurethral zone

- Subregion of transitional zone
- Mucosal + submucosal glands

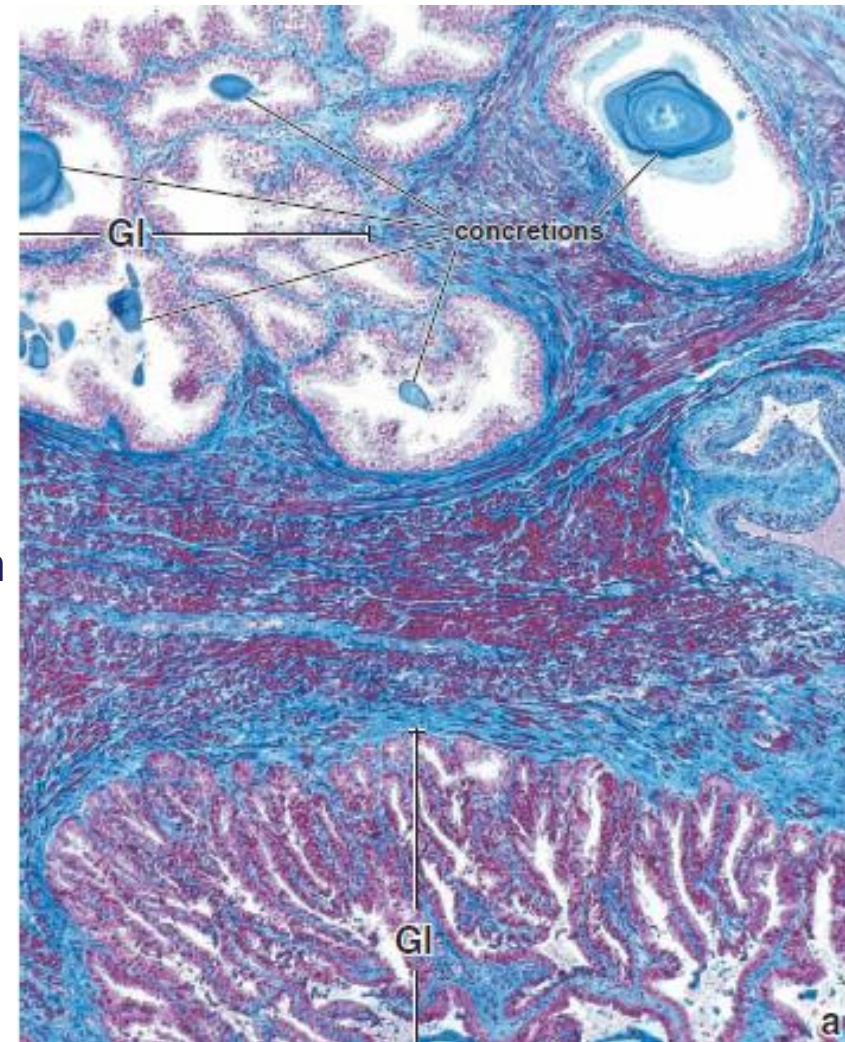


Accessory Sex Gland: Prostate Gland

Secretes a clear (watery), slightly alkaline (pH 7.29) fluid into the prostatic urethra

- **Prostate-specific antigen (PSA)**
 - Serine protease
 - Tumor biomarker
- **Prostatic acid phosphatase (PAP)**
 - Regulates cell growth + metabolism
 - Elevated in metastatic prostate ca.
- **Fibrinolysin** – liquifies semen
- **Citric acid** – nutrient for sperm

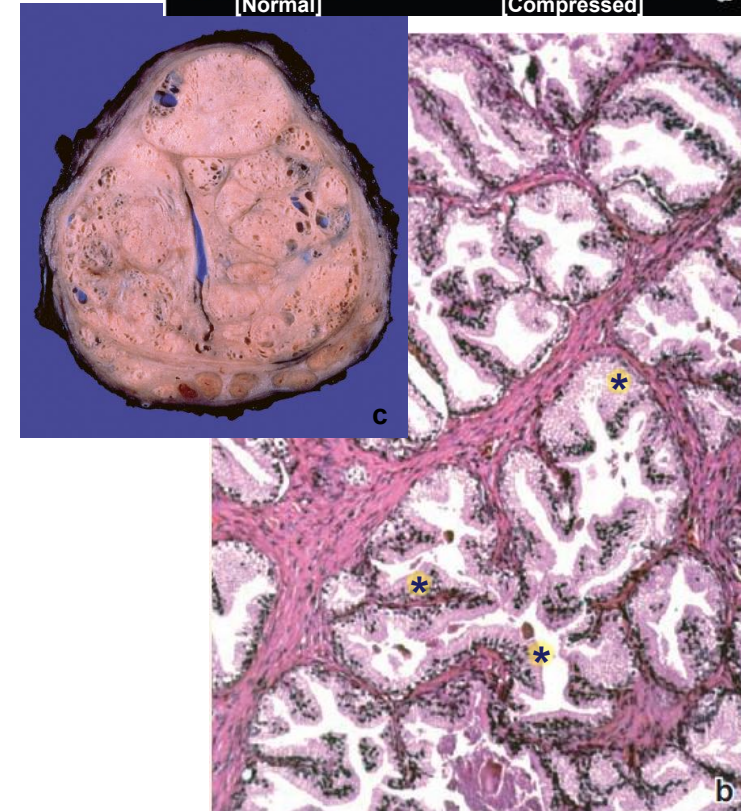
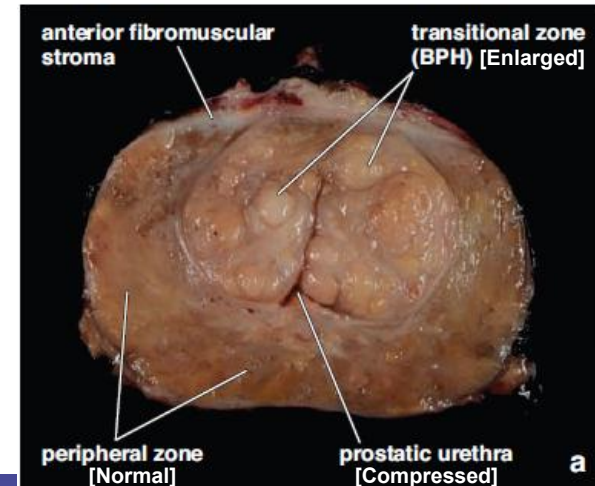
Produces ~25-30% volume of semen



GI - glands

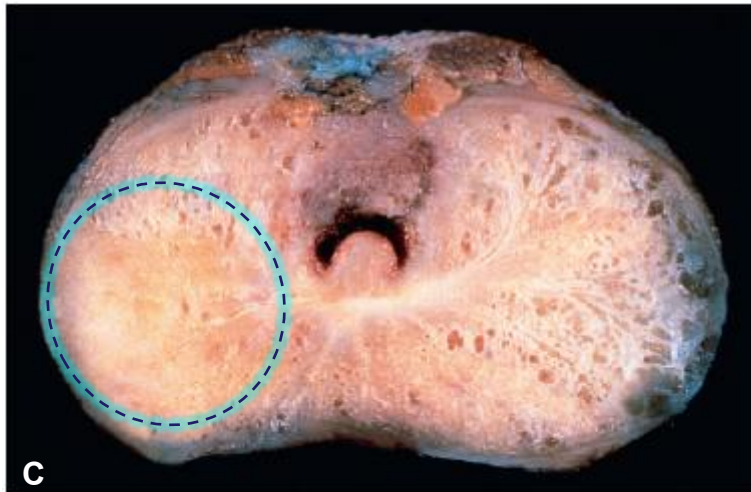
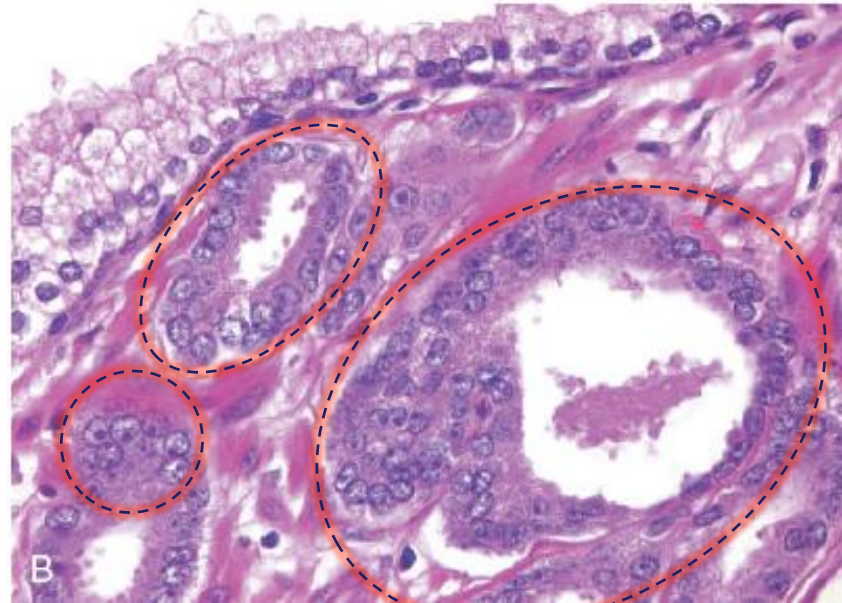
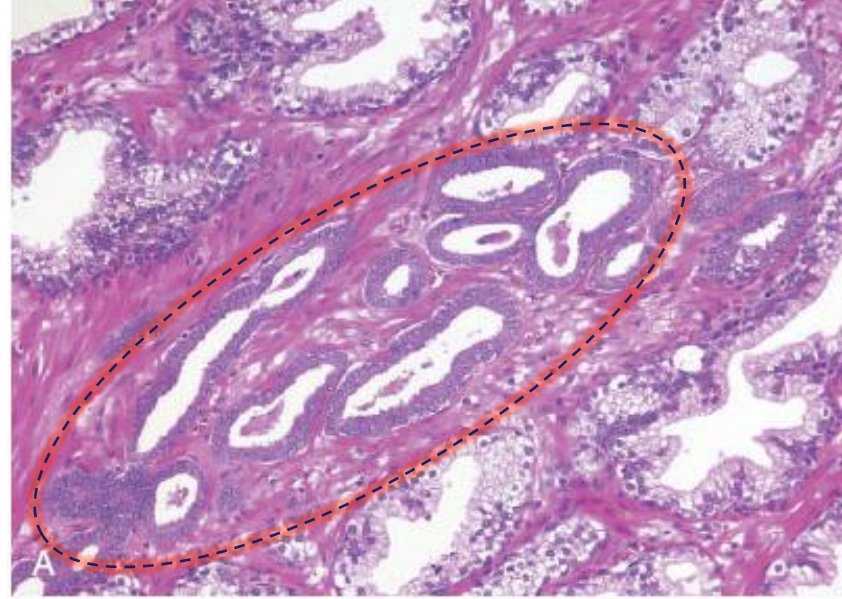
Prostate Gland: Benign Prostatic Hyperplasia (BPH)

- **Proliferation of glandular epithelium**
- **Hyperplasia (*b*) → nodule formation**
 - **Role of dihydrotestosterone (DHT)**
 - 5 α -reductase: Testosterone → DHT
 - Stimulates epithelial growth
 - Cells fold + protrude into lumen (*)
 - Fibromuscular stromal growth
 - **Enlarged prostate gland**
- **Macroscopic findings (a,c):**
 - Well-circumscribed, bulging nodules
 - Solid or cystic spaces (dilated glands)
 - Compressed, slit-like prostatic urethra
- **Almost exclusively – transitional zone**
 - Obstructs prostatic urethra
 - Weak urination, nocturia
 - Difficulty starting (hesitancy) & stopping
 - Urge or feeling to urinate (urgency)
- Believed to occur to some extent in all males by age 80



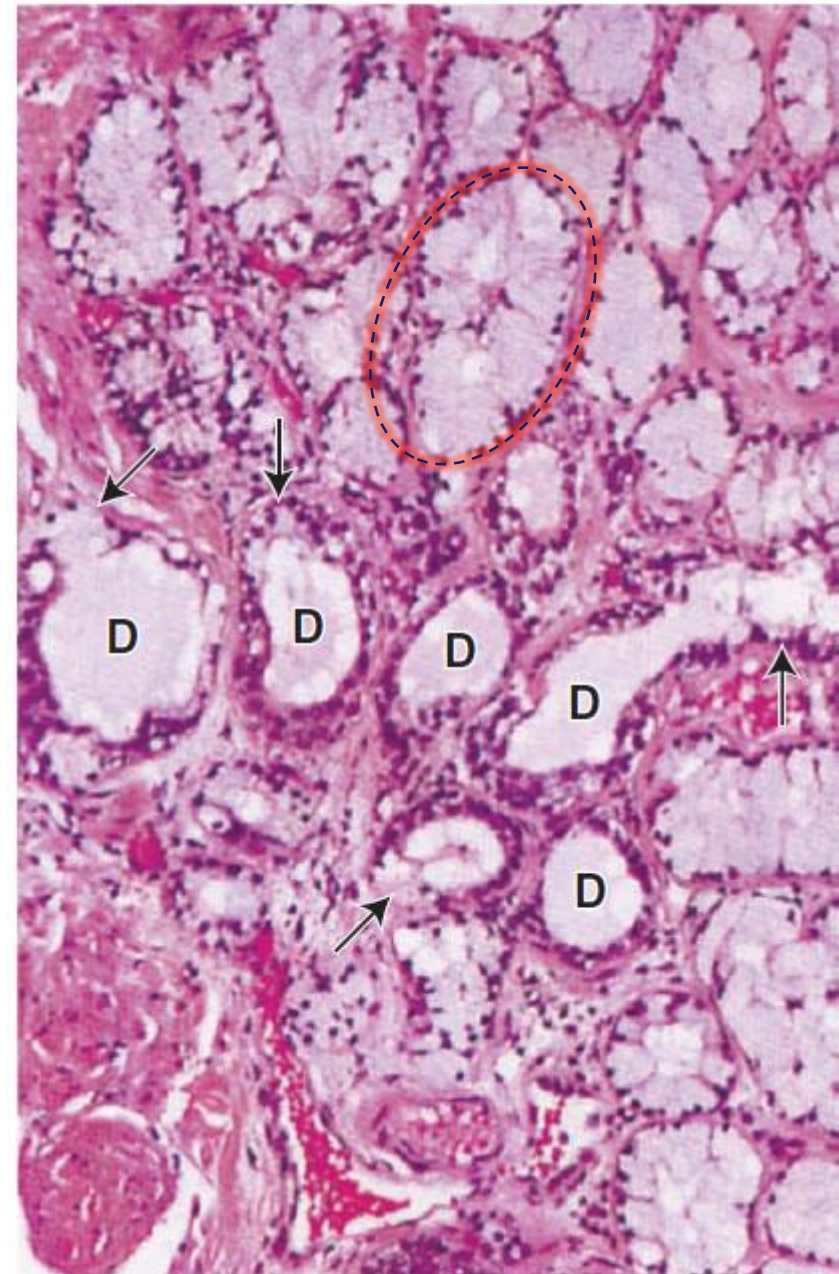
Prostate Gland: Adenocarcinoma

- Most common form of cancer in males
- Typically affects men over 50 years
- **Macroscopic findings (C)**
 - Arises in peripheral zone
 - Firm, solid, grey-white, ill-defined lesion (*advanced disease*)
- **Microscopic findings**
 - Well-defined glandular pattern
 - Small, round glands crowded between larger benign glands (A)
 - Cuboidal cells, dark cytoplasm, enlarged nuclei, prominent nucleoli (B)
 - Lacking basal layer; overcrowded



Accessory Sex Gland: Bulbourethral Gland

- Located in urogenital diaphragm
- **Compound tubuloalveolar glands**
 - Resembles mucus secretory glands
- Duct (*D*) penetrates inferior fascia of urogenital diaphragm
 - Joins + empties into penile urethra
- **Glandular epithelium**
 - Simple columnar epithelium
 - Mucus secreting cells
 - *Basally located nuclei*
 - *Clear apical cytoplasm*
 - *Also line ducts (arrows)*
 - Varies based on functional state (testosterone)
- **Clear, mucus-like secretions**
 - Preseminal fluid
 - Lubricates urethra
 - Neutralizes acidic urine





Summary Table

	Seminal Vesicle	Prostate Gland	Bulbourethral Gland
Paired or Unpaired			
Anatomical Shape + Location			
Location where duct drains (*duct formation)			
General gland Features			
Glandular Epithelium			
Smooth Muscle			N/A
Fibrous coat			N/A
Secretion properties (e.g., color, consistency, pH)			
Secretory components (function)			
Volume of Semen			
Hormonal Influence			

Thank you

If you have any questions, kindly send it to
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